

Cloud Computing Tutorial - 3

Create a Virtual Machine Using VMware Workstation Pro 12 and Install Python

Observation:

Session 3

1) Creating a Virtual Machine

- * VMware environment / workstation Pro 12 was installed and launched.
- * A new virtual machine was created using the custom option.
- * Installation model was selected using an ISO file.
- * The guest operating system and its version were chosen.
- * The virtual machine was configured by allocating RAM, CPU core and disk space.
- * The set up was completed and operating system was installed successfully.

2) Installing Python on the Virtual Machine

- * The VM was started and the user logged into the operating system.
- * For Linux: Python and pip were installed using terminal commands.

For Windows: Python installer was downloaded.

3) Running a simple Python Program

- * A python script was created to implement a basic machine learning model inside the Virtual Machine.
- * Required libraries such as NumPy, Pandas, scikit-learn was imported for data handling and model building.
- * A dataset was loaded and preprocessed by handling missing values and selecting relevant features.
- * The dataset was divided into training and testing sets to evaluate the model performance.

- * A suitable machine learning algorithm such as :
 - Linear Regression
 - Logistic Regressionwas selected and trained using the training data.

- * The trained model was then tested using the test dataset to make the predictions.

- * The performance of the model was evaluated using appropriate metrics such as accuracy, mean squared error or precision.

- * The program was executed successfully and the output confirmed that the Machine Learning model was correctly working within the virtual environment.

Oracle Virtual Machine demonstration:



Fig 1.1: Oracle VirtualBox Installation

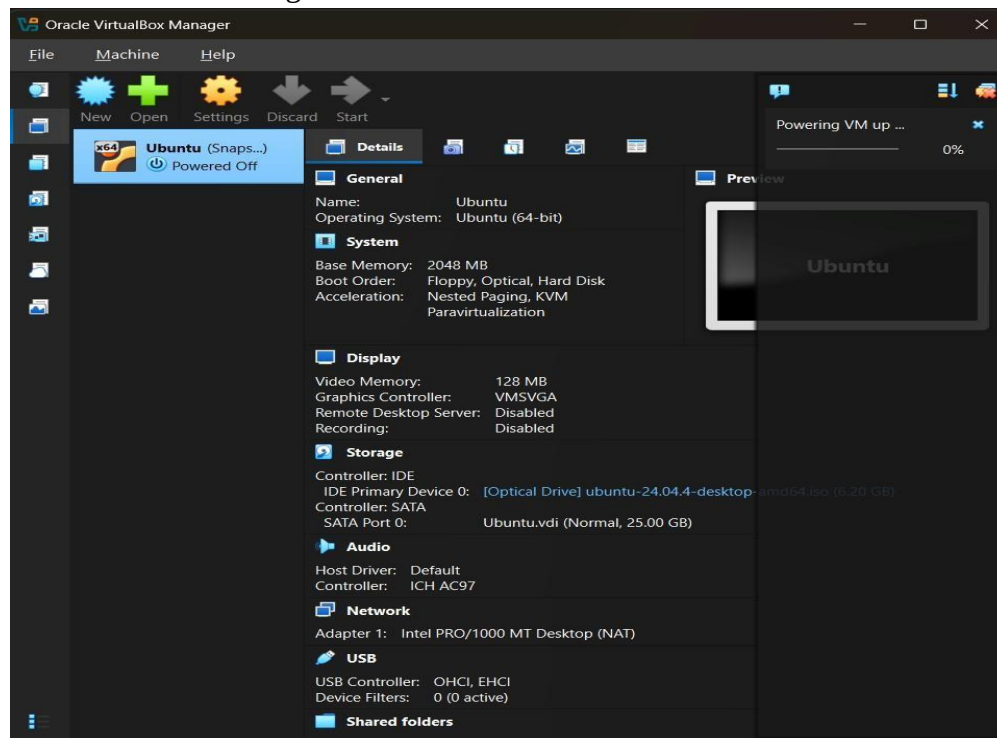


Fig 1.2: Turning on Ubuntu virtual machine on Oracle VirtualBox Manager

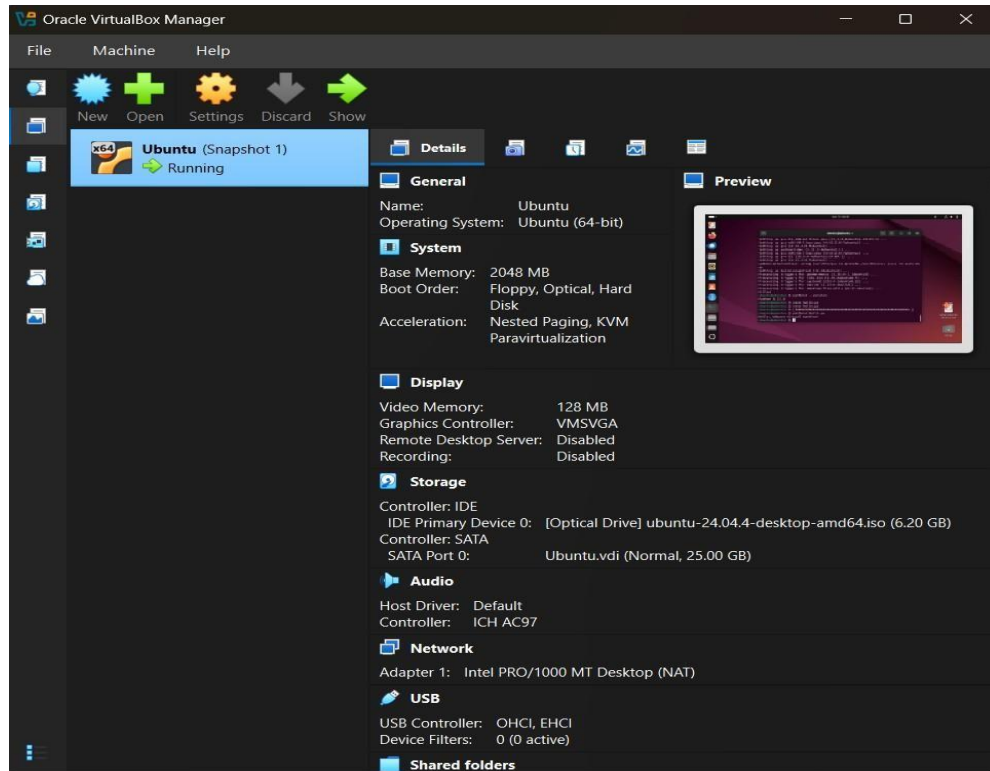


Fig 1.3: Ubuntu virtual machine running

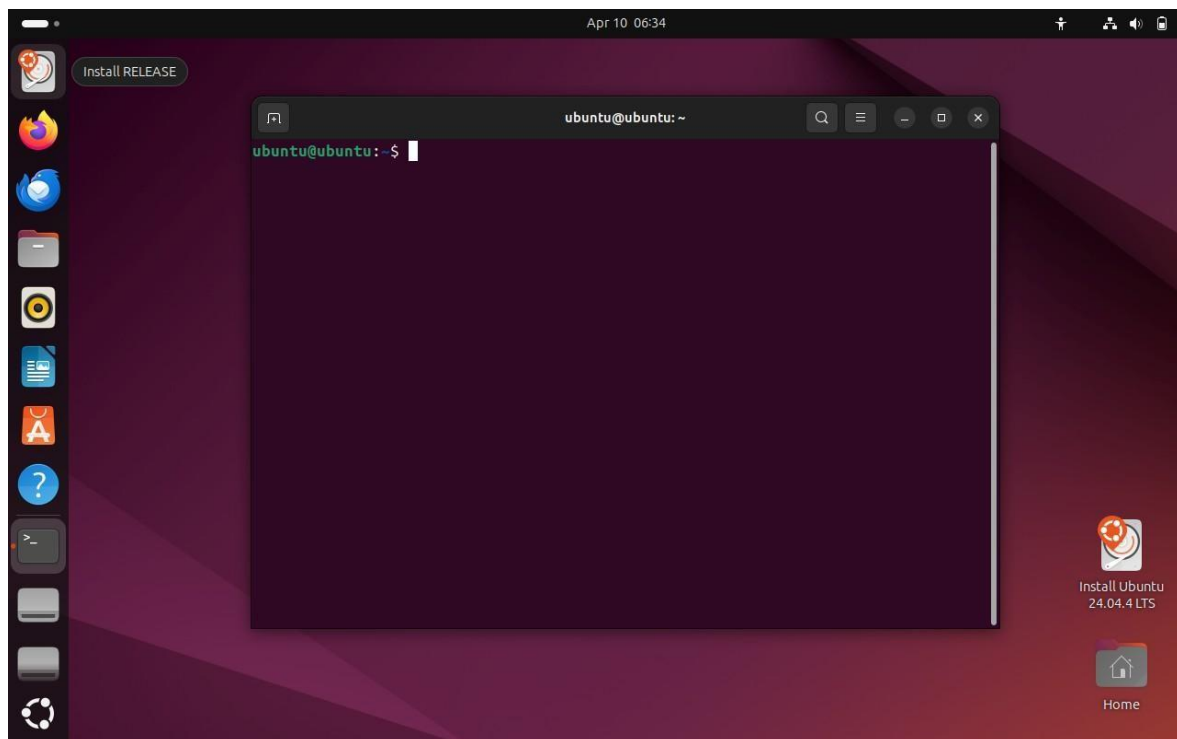


Fig 1.4: Ubuntu Terminal

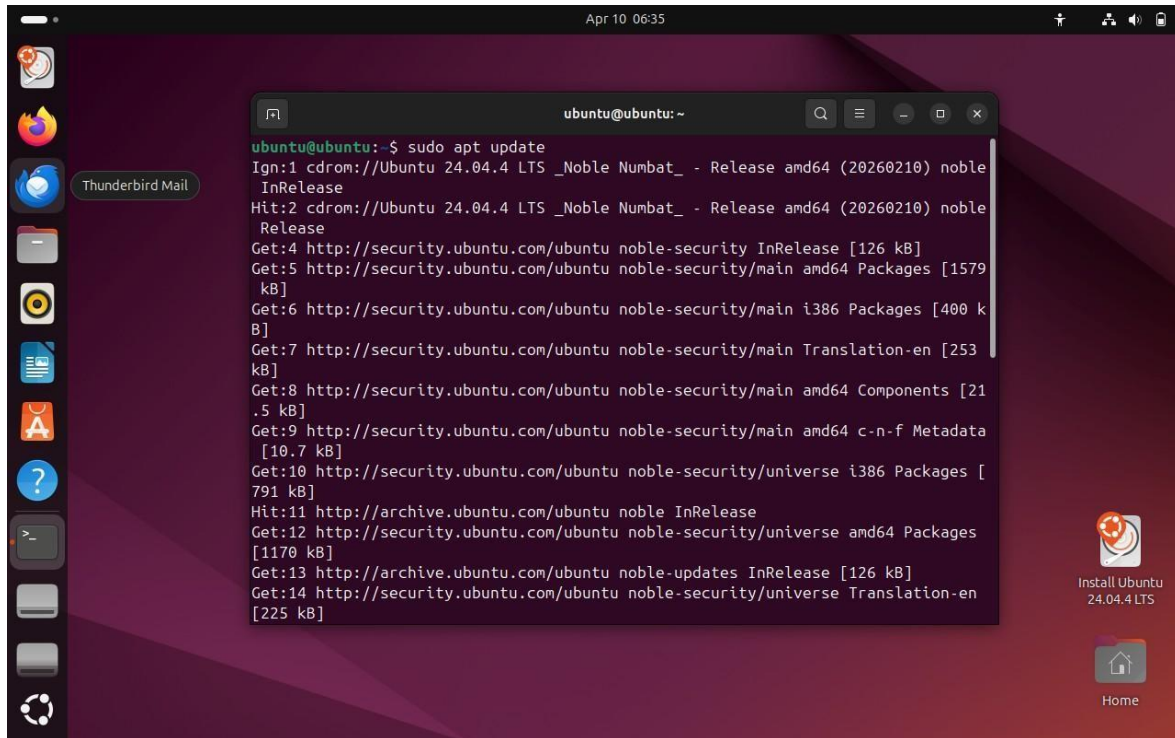


Fig 1.5: Updating package manager

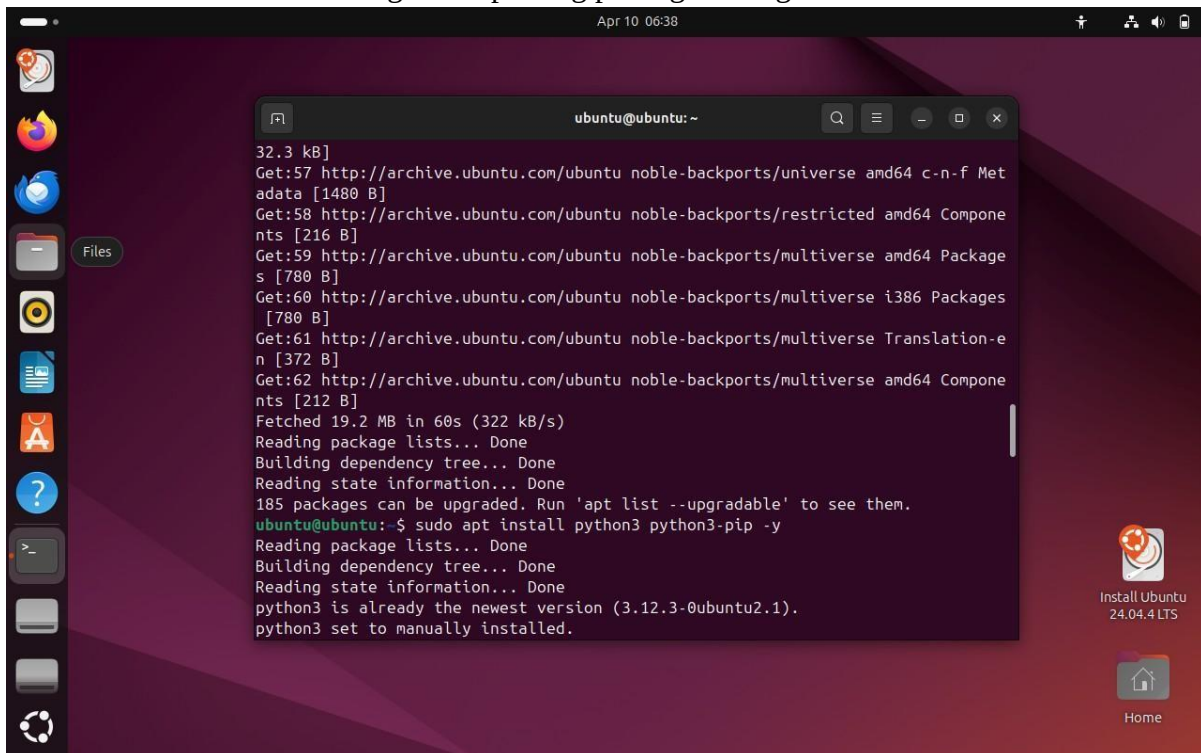


Fig 1.6: Installing python

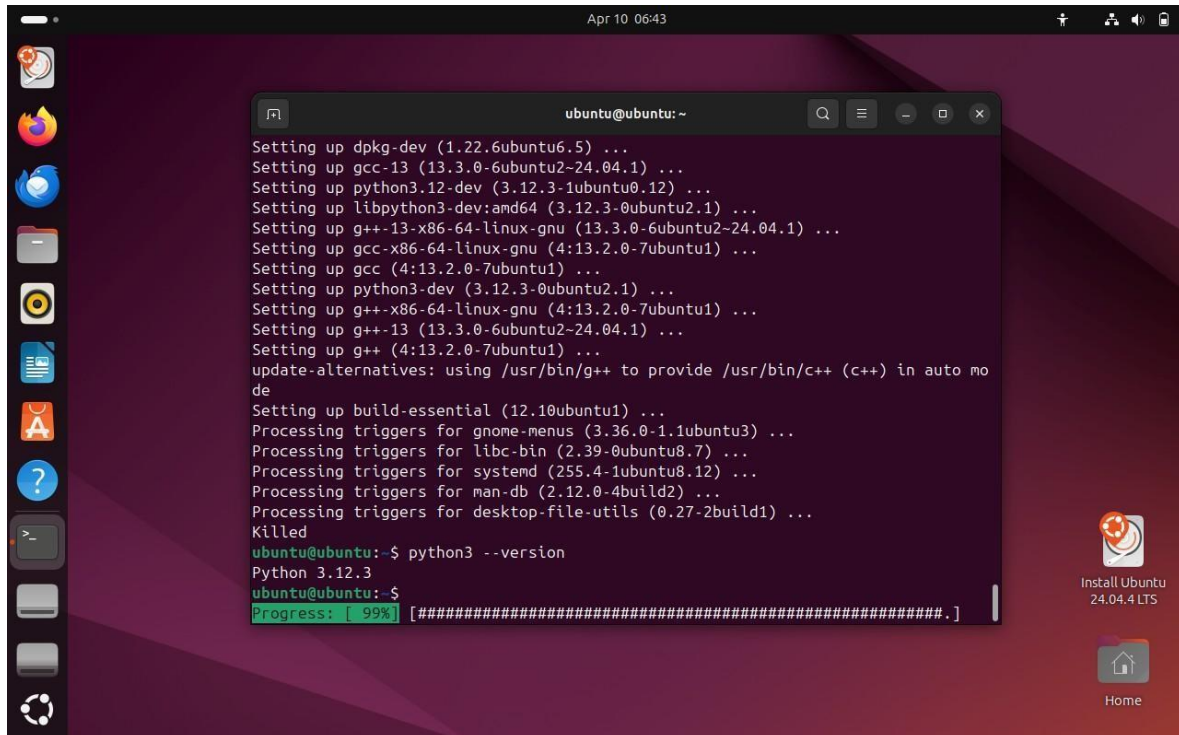


Fig 1.7: Installation successful

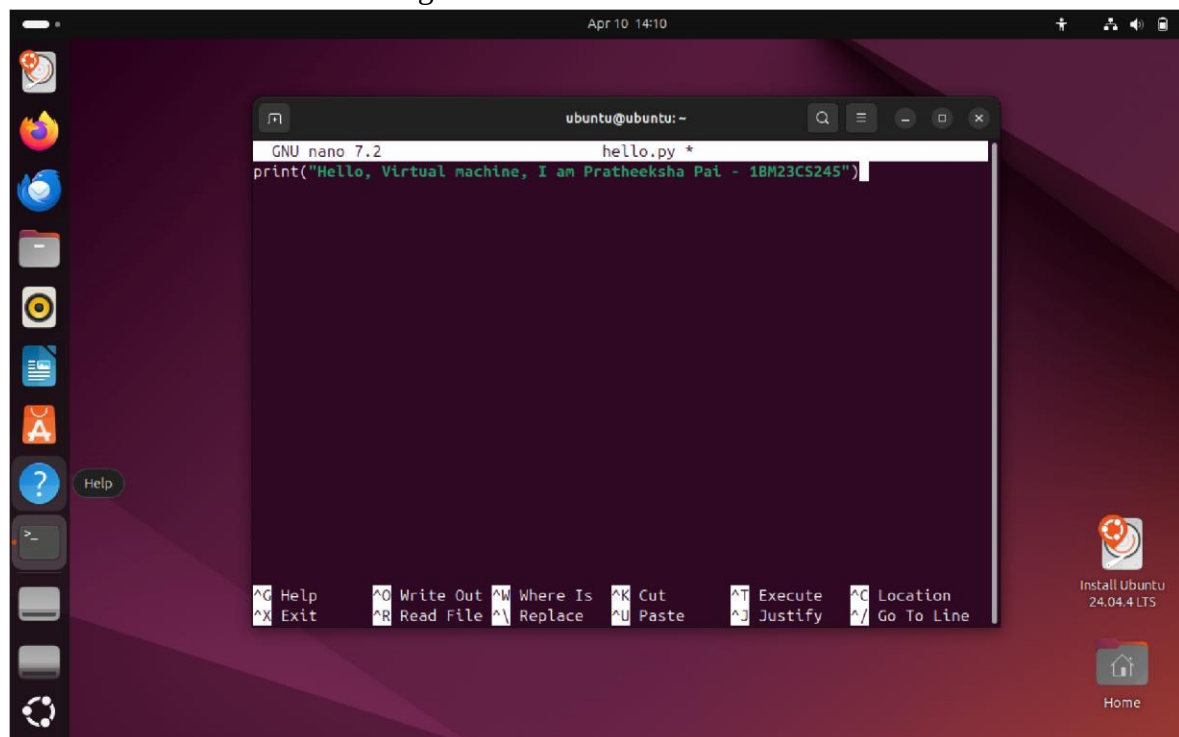


Fig 1.8: Python code

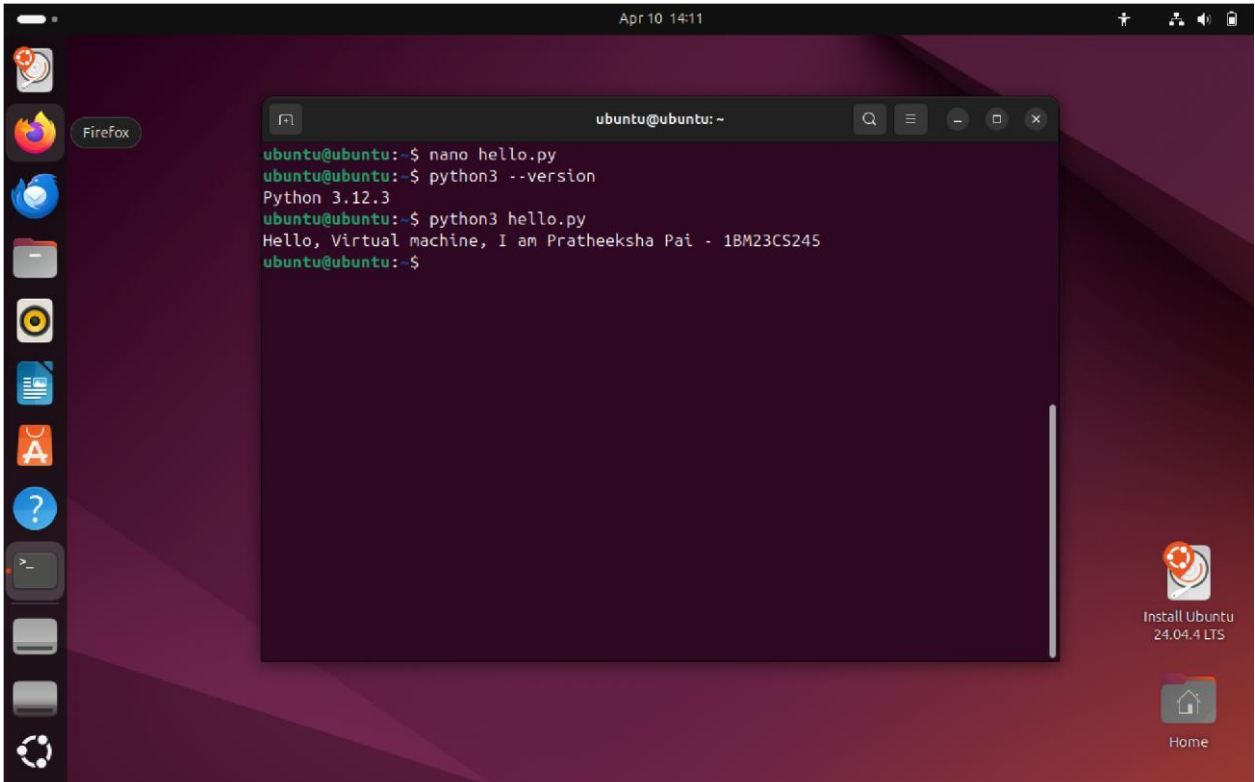


Fig 1.9: Final python program output